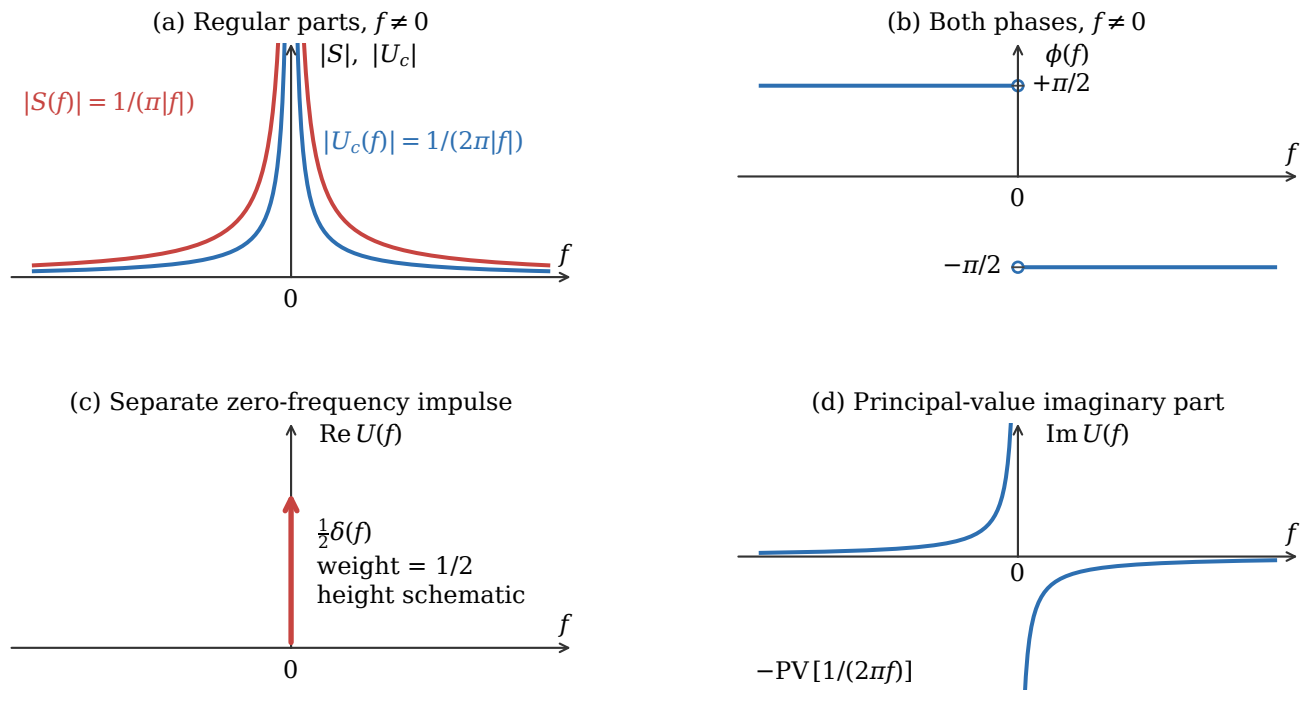
(a) Regular parts, $f \neq 0$
$|S|,\ |U_c|$
$|S(f)| = 1/(\pi|f|)$
$|U_c(f)| = 1/(2\pi|f|)$
f
0
(b) Both phases, $f \neq 0$
$\phi(f)$
$+\pi/2$
f
0
$-\pi/2$
(c) Separate zero-frequency impulse
$\mathrm{Re}\,U(f)$
$\frac{1}{2}\delta(f)$
weight = 1/2
height schematic
f
0
(d) Principal-value imaginary part
$\mathrm{Im}\,U(f)$
f
0
$-\mathrm{PV}[1/(2\pi f)]$